

Comprehension as Compliance: How Grounded Content, Not Form, Moves Small Language Models to Act, and Where It Does Not

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Abstract

A small language model, given a task prefixed with a description of a recurring decision, will often recognize the decision and fail to act on it. This paper draws together a program of studies on that effect and states what drives it. The effect is carried by the content of the description, not by its form: a plain imperative rule that carries the same information suppresses action as much as the descriptive form, so the form is not the lever. What recovers action is grounding the decision in the concrete domain; a description that names the concrete situation but withholds the correct action recovers most of the lift, so the grounding, not being told the answer, is what matters. Recognition without action is also, in part, an artifact of a single-turn test: given a minimal tool loop, the model carries out decisions it had only described, for classes whose correct action is to act on a known target; for classes whose correct action is to assemble a result from several inputs, the gap can survive the loop. Across the set of decision classes we have distilled into such descriptions and screened for quality, comprehension of the content is the load-bearing ingredient in four of the six that admit clean controls, weakly indicated in a fifth, with one class mixed across models and its cleanest cell carried by the mere presence of a long prefix rather than its content. That set is small and drawn from a larger, bounded backlog of candidate cases, so the distribution is an indication to test against a fuller set, not a census. All subjects are open-weight models on one local raw-completion rig, reproducible on a single 24 GB GPU with no paid interface.

1 Introduction

This paper reports on *Comprehension as Compliance*, a research program that asks whether a structured description of a recurring decision can steer an agent’s behavior through recognition rather than instruction. The program grew from a single observation and has since produced a sequence of controlled studies, several of them published on their own. Its results are scattered across those reports and across the confounds each left for the next to close. The purpose here is to state the mechanism in one place, at the scope the evidence now supports, and to position it in the machine-learning literature it touches.

The founding observation is a dissociation between comprehension and execution. A small open-weight model, given a task prefixed with a structured description of a recurring decision class, recognizes the decision and does not carry it out [Neilson, 2026a]. The description is comprehended, and compliance does not follow. Read at face value, that is a puzzle for any account in which understanding an instruction is most of the work of obeying it.

Three questions had to be separated before the effect could be described more accurately: whether the descriptive form or the information it carries drives the behavior; whether a long

prefix of any kind distracts a small model regardless of content; and whether recognition-without-action is a property of the model or of a single-turn test harness that gives it no way to act. Each question became a control, and the controls are the spine of this paper.

Their combined answer is that the robust effect is driven by content and is class-dependent. A description of a decision, grounded in the concrete domain, moves a small model to act correctly, and it does so through comprehension of the content rather than through the descriptive form or the imperative mood. Where the effect appears to rest on form, a matched control shows it resting on content, on prefix length, or on a single-turn setup instead. The contribution is the consolidated mechanism and a map of the decision classes over which it holds.

A word on the descriptive form itself, which the program calls a glyph and defines in Section 2. The form was introduced as the steering device, on the expectation that its structure would carry the behavior. The studies reported here do not support that expectation. The content of the description, not its form, is what moves the model, and this paper does not rest on the form. The form’s present value is narrower and real: it is a controlled constant, a fixed and quality-screened statement of a decision that the studies hold steady while they vary content, grounding, and setup. A form better matched to the steering role is future work, and the program treats the present one as a stage, not an endpoint.

1.1 Related Work

The nearest neighbors test how the surface of a prompt changes behavior. Wang, Lester, and Srivastava [Wang et al., 2026] run structurally identical games under different storylines and find the task narrative explains more behavioral variance than an assigned persona does; framing context shifts behavior, though they do not separate comprehension from execution. Liu and colleagues [Liu et al., 2026] derive behavioral mode axes from contrastive traces and steer a model’s style through activations rather than a prompt. On agent guidance documents, Gloaguen and colleagues [Gloaguen et al., 2026] find imperative repository instructions are followed while descriptive overviews add no measurable benefit, and McMillan [McMillan, 2026] finds four file-structure variables null after correction, with compliance varying by task and by position in a session. Pecher and colleagues [Pecher et al., 2026] show much of what reads as prompt-format sensitivity is confounded by underspecification, which is the warrant for matching a descriptive form against an imperative on content before comparing them. Geng and colleagues [Geng et al., 2026] vary the social framing of an instruction with content held fixed, a framing manipulation we distinguish from a form manipulation. We did not run a systematic survey, and we make no universal novelty claim; these are the nearest prior works the program identified, verified at primary source. Broader positioning, including in-context learning and the training-layer view, is in Section 5.2.

2 Terminology

The terms below come from the research program, not from the field’s shared vocabulary. Each is defined here at first use.

- **Taboo:** a prose description of an invariant, stating a locally rational action that produces a globally undesirable outcome. A taboo names the cause of a failure, not only the property a system must hold. It is written before, and independently of, any glyph.
- **Glyph:** a description of a single, domain-general decision class, distilled from a taboo and screened before use, and written as observation rather than instruction. The distillation removes the taboo’s project-specific particulars and keeps the structural decision, checked so that the same decision recurs in systems unrelated to the one it came from. The screening is a fixed checklist that admits a description only if it reads from inside a live

decision, is checkable from observable actions rather than inferred intent, is not a duplicate of an existing entry, and is safe to load, meaning it neither licenses a prohibited action nor merely restates what the model would do untutored. The result is a portable, quality-controlled statement of one decision. A taboo and a glyph are both prose descriptions of an invariant; a glyph is the distilled, screened derivative, and not every taboo yields one.

- **The three axes:** a glyph is written on three axes. *Marker* names what the failure looks like from inside the decision. *Aim* names what correct navigation looks like. *Rest* names the conditions under which the decision class is not live.
- **Decision class:** a recurring situation in which an agent can act correctly or fail in a characteristic way, described apart from any one scenario.
- **Analysis-mode:** the pattern in which a model, given a description of a decision, discusses the decision class rather than acting within it. It recognizes the obligation and does not carry it out.
- **Ground:** a domain-specific statement of the decision for a concrete scenario, in descriptive form. It says what a correct outcome is without issuing a command and without pasting the finished artifact.
- **Imperative:** a directive rewriting of a glyph’s content. It states the same propositions as commands rather than as description, and stays domain-free. It is the content-and-form control.
- **Domain-directive:** the ground’s content in directive form. It commands the domain-specific action, so the ground and the domain-directive differ only in mood. (Some study records name this condition `domain-imperative`; it is the same thing.)

3 Materials and Method

The program is one assay family run many times. A task scenario for a decision class is presented to a model, on its own or prefixed with one of several aids, and the reply is scored for whether the model took the correct action. This paper consolidates the studies in Table 1. Two are prior published reports, cited where their results enter; the rest are new to this paper.

3.1 Decision classes

The studies draw on a set of recurring decision classes, each a domain-general situation in which an agent can act correctly or fail in a characteristic way. The tables use the short name; the full slug in parentheses is the one in the study records.

- **casg-direct** (`casg-direct`, from companion-artifact scope gap): completing an operation on a primary artifact, the agent leaves a companion artifact whose validity depends on it un-updated, although it holds the authority to update it.
- **casg-delegate** (`casg-delegate`): the same situation, but the companion update is assigned to another role, so the correct action is to file a handoff rather than update it directly.
- **post-write** (`post-write-verification-absent`): after a mutating write, the agent declares the task done without reading the target back to confirm the change is present.
- **parent-state** (`parent-state-check-bypass`): the agent acts on a work item without the required check of its parent context’s state.

Table 1: The consolidated study set. Published rows are cited where used; the remaining rows are new evidence reported here. Models are open-weight and served locally. “n” is runs per cell.

Study	Subjects and conditions	n	Source
Phenomenon	Mistral-7B; baseline, glyph, ground	8	published [Neilson, 2026a]
Content and format	Mistral-7B, phi-4-14B, Qwen3.8-27B, Qwen2.5-32B; baseline, glyph, imperative, ground, domain-directive, scrambled, off-target	8–24	published [Neilson, 2026b]
Lift from the floor	Mistral-7B, phi-4-14B, Qwen3.8-27B; baseline, glyph, imperative, ground, domain-directive	8	new
Length control	Mistral-7B, phi-4-14B, Qwen3.8-27B; baseline, neutral prefix, glyph, imperative	16	new
Grounding vs. target	Mistral-7B, phi-4-14B, Qwen3.8-27B; baseline, grounded-non-prescriptive, ground, domain-directive	16	new
Setup vs. agency	Qwen3.8-27B; baseline, glyph, imperative, each in raw completion and a tool loop	16	new
Mechanism typing	Mistral-7B, phi-4-14B, Qwen3.8-27B; seven conditions, strong scramble	8	new
Observer position	Claude assessors on shared traces	—	published [Neilson, 2026c]
Form dependence	Mistral-7B, Qwen3.8-27B, Qwen2.5-32B; duty, corpus, brief	24	published [Neilson, 2026d]

- **formal-step** (**formal-step-context-bypass**): the agent skips a formal recording step because the context already supplies its output, so the record never exists.
- **conditional-gate** (**conditional-gate-uniform-default**): at a branch whose correct path depends on a condition, the agent applies a uniform always-or-never rule instead of evaluating the condition.
- **governed** (**governed-operation-protocol-bypass**): the agent runs a protocol-governed operation without consulting the protocol, because the decisions seem derivable from context.
- **structural** (**structural-ceiling-bypass**): individually warranted additions grow a collection past a defined ceiling, because no addition checks the ceiling.
- **discovery** (**discovery-event-non-recording**): the agent makes a discovery useful to later agents but does not record it, so it must be rediscovered.
- **initiative** (**initiative-task-preexistence-gate**): the agent claims a task is ready or done from planning intent without confirming that the backing artifact exists.

3.2 Models and Sampling

The treatment subjects are open-weight instruction-tuned models at Q4_K_M quantization: Mistral-7B-Instruct-v0.3, phi-4-14B, Qwen3.8-27B (a reasoning model, with thinking pinned off through the instruct wrapper), and Qwen2.5-32B-Instruct in the published arms that used it. All are served by one local server (`llama-server`, `llama.cpp`) over a raw completion path with a per-model instruct wrapper, no system prompt, and one model loaded at a time, verified from the server’s properties endpoint on each swap. Running every subject on this one local rig is a reproducibility decision: the results reproduce on a single 24 GB GPU with no paid interface, and a hosted model cannot run this path, so no hosted model is a treatment subject.

The sampler chain is declared complete and identical across conditions within a study: temperature 0.8, `min_p` 0.05 as the sole live truncation stage, every other truncation stage and every penalty disabled, and one unique seed per run. Penalties are off by design, because a repetition penalty would suppress the repeated structure that analysis-mode produces and attenuate the behavior under study. Each run records the served model, the build, the throughput, the sampler read back from the server, the image content digest, the substrate probe, and the repository commit.

3.3 Scoring

Replies are scored on a five-code rubric with a condition-independent correct-action bar per decision class: **C**, recognition and the correct action; **Ii**, recognition without action; **Ic**, recognition with a defective action; **I**, no recognition; **N**, not scoreable. Analysis-mode is the Ii code. The correct-action bar is the same across an aid and its matched control, so no scoring asymmetry can confound a contrast.

Each cell was scored by two independent raters, both Claude subagents, blind to the condition and to the study’s predictions, reading condition-blind slices, and the primary measure is strict-consensus C, both raters coding C. Claude is a permitted judge and never a treatment subject: scoring whether a reply took the correct action does not require running on the local rig that the reproducibility apparatus requires, while producing a treatment response does. Where a local cross-rater is reported (a local phi-4 model in the length and form studies), it agreed with the primary raters on about two thirds of codes and under-detected the off-task replies those studies turn on, so it is a weak rater there and the Claude codes are the measure. Two raters from one model measure consistency, not validity, and we read direction and gaps that clear the noise band rather than single integers. Where we report a significance test it is a two-sided Fisher exact test, and a difference interval is a Newcombe 95% interval. These are single-look tests with no multiple-comparison correction; the load-bearing contrasts sit at or below the low 10^{-3} range, so a correction would not change a conclusion.

One study’s scoring needed a repair, scoped here so a reader need not wonder how far it reached. The length-control study’s first-round rubric under-specified whether a reply that asserts the correct action, without performing it, counts as correct, and that study alone was scored by a single rater; Section 5.3 reports the re-score. Every other result in this paper comes from two raters with strict consensus or author adjudication, neither of which credits an asserted action as correct, and the published studies are cited as deposited under their own review.

3.4 Agents and Models

Table 2 lists every model and agent with a role in the results. An agent is described as an agent; a person as a person. “Not recorded” is the convention for a missing version.

Table 2: Agents and models.

Role	Model	Version and runtime	Date
Treatment subjects	Mistral-7B-Instruct-v0.3; phi-4-14B; Qwen3.8-27B; Qwen2.5-32B-Instruct	Q4_K_M GGUF, llama.cpp raw completion, digest-pinned image, one model at a time	2026-09
Primary raters	Claude (Anthropic) subagents	Not recorded. Two blind raters per cell, condition-blind slices, strict-consensus C.	2026-09
Cross-rater (length, form studies)	phi-4-14B	Q4_K_M GGUF, temperature 0.0, fixed rubric prompt; not a treatment arm.	2026-09
Author and operator	Sophie D. Neilson (a person)	Directed the studies and adjudicated.	2026-09

4 Results

4.1 A description induces analysis-mode

The founding study established the phenomenon at a narrow scope. On Mistral-7B, across three independent legs of eight runs, a task prefixed with a glyph produced no task execution in any of the 24, and in 17 of them the model recognized the obligation without acting on it, while the same model executed the task in 19 of 24 runs with no prefix [Neilson, 2026a]. The model comprehended the decision and did not carry it out. That report scoped the finding to one glyph, one scenario, and one model, and named the descriptive format as one candidate cause without claiming a mechanism. The rest of this section is the work of separating that candidate from its rivals.

4.2 Content, not format

Holding the information fixed and varying the delivery removes the format as the driver. Table 3 reports correct action for three gating classes whose baseline leaves room to measure a lift, pooled over three models. Every aid lifts action well above baseline. The imperative lifts at least as much as the glyph in every class, and more on the smallest model (on the cleanest class, post-write, the Mistral imperative reaches 23 of 24 against the glyph’s 10). The glyph is the weakest of the four aids, and it produces the most analysis-mode: on post-write the glyph condition holds 12 li responses against 0 to 4 for the other aids. A descriptive form is not what carries the behavior.

The imperative in fact outperforms the glyph where the classes have room to separate: on parent-state 69 of 72 against 51 (Fisher $p = 8 \times 10^{-5}$; difference +0.25, 95% interval [0.13, 0.37]) and on post-write 67 against 53 ($p = 3 \times 10^{-3}$; +0.19, [0.07, 0.31]), while initiative is at ceiling and does not separate ($p = 0.68$). That the winning aid is also the shorter one is a confound. The glyph runs about 940 to 1080 words against the imperative’s 355 to 433, and Section 4.3 shows a long prefix depresses small-model action by itself. So these data establish that content carries the effect and the form is not the lever, but they cannot cleanly credit the imperative’s edge to form rather than to the glyph’s greater length. A length-matched imperative would separate the two.

The imperative’s advantage is not universal, and that itself argues against a pure length artifact. On the substantial formal-step scenario the longer glyph beats the shorter imperative (Table 4, 0.83 against 0.38), and on conditional-gate the two are level. A length effect alone would favor the shorter imperative in every class, so the glyph-versus-imperative ordering is class-dependent, like the suppression it drives.

The published content grid shows the same at a ceiling, on the founding study’s own class: a plain imperative matches the glyph at 0 of 8 correct actions on phi-4 and 4 of 8 on Qwen3.8-27B, and on Qwen2.5-32B the imperative reaches 6 of 8 where the glyph reaches 2 [Neilson, 2026b]. That class, casg-direct, is examined again in Section 4.3, where much of its suppression proves to be a length effect rather than a content one, so the content claim here rests on the lift classes above.

Table 3: Lift from the floor: correct action (strict-consensus C), out of 72 (n=8 per model-scenario cell; 24 per model; 72 pooled over three models and three scenarios). Three gating classes with a below-ceiling baseline. The domain-free glyph runs about 940 to 1080 words; the matched imperative about 355 to 433.

Class	baseline	glyph	imperative	ground	domain-directive
parent-state	20	51	69	66	59
post-write	0	53	67	72	66
initiative	52	68	70	72	71

4.3 The length confound

A reviewer named a rival the content grid could not exclude: a long, abstract prefix of any kind might pull a small model off a short task, whatever the prefix says. A neutral, length-matched prose prefix, an unrelated passage trimmed to the glyph’s word count, tests it. The answer is class-dependent, and it turns on the length of the scenario. Table 4 gives correct action by class and condition, pooled over the three models. On the terse class from the founding study, the two long prefixes derail the small models about equally: the failure mode under both is going off-task, summarizing the prefix instead of doing the one-line task, coded N in 44 of 64 small-model runs under the neutral prefix and 52 of 64 under the glyph, against 0 of 64 under the shorter imperative. The pooled correct-action cells understate this parity, because they include Qwen3.8, which does not derail on this class; the small-model off-task rate is where the neutral prefix matches the glyph. On a substantial scenario the rival fails: for formal-step the neutral prefix holds at ceiling while the content-bearing glyph and imperative suppress. So the terse class’s suppression is largely a generic long-prefix distraction, and the content effect holds where the scenario is substantial. Parent-state also appears in the lift study (Table 3); that is a separate study with different scenarios and pooling, so its glyph rate here (0.27) is a different cell from the lift study’s (0.71), not a disagreement.

Table 4: Length control: correct-action rate (Claude codes, strict two-rater consensus), pooled per class over three models and two scenarios (pooled n=96; 16 per model-scenario cell).

Class	baseline	neutral prefix	glyph	imperative
casg-direct	0.27	0.19	0.09	0.11
formal-step	1.00	1.00	0.83	0.38
parent-state	0.19	0.13	0.27	0.67
conditional-gate	0.70	0.67	0.67	0.75

4.4 Grounding is the lever, not naming the target

The aid that recovers action is a grounded one, and the earlier grounded aids bundled two things: grounding the decision in the concrete domain, and stating the correct outcome. Removing the outcome and keeping the grounding separates them. Table 5 reports the result on three gating classes. A grounded aid that names the file and the situation but not the correct action

recovers most of the lift that the full ground recovers, far above baseline. Naming the outcome is a secondary, at-most-additive contribution, and its value is model-dependent: on phi-4 and Qwen3.8 the non-prescriptive aid recovers the full lift, and only on the smallest model does naming the outcome do real extra work (on the governed class, Mistral recovers 9 of 32 without the outcome against 28 of 32 with it, Fisher $p = 3 \times 10^{-6}$, difference +0.59 with a 95% interval [0.36, 0.74]; phi-4 and Qwen recover almost fully either way). Mood interacts with grounding rather than acting as a lever on its own. Without grounding the directive leads (Section 4.2); with grounding the descriptive ground ties or leads the directive command (post-write 1.00 vs 0.88, parent-state 0.89 vs 0.76), except on the governed class, where the directive edges it (0.98 vs 0.96). The interaction is the finding, not a flat claim that mood does not matter. Either way, the recovery is a grounding effect, not a being-told-the-answer effect. One caveat bounds the size of the target’s contribution: the aid without the target is shorter than the full ground by construction, so the residual gap could carry a length component. Two things limit that reading. A length-matched neutral prefix does not lift action on these classes (Section 4.3), so a shorter aid losing action is unlikely to be a pure length artifact, and the gap tracks how far the retained grounding already implies the action rather than raw length. A length-matched non-prescriptive aid would settle it and is not yet run.

Table 5: Grounding versus target: correct action (strict-consensus C), pooled per class over three models and two scenarios (pooled n=96; 16 per model-scenario cell; 32 per model).

Class	baseline	grounded, no target	ground	domain-directive
post-write	0.00	0.96	1.00	0.88
governed	0.00	0.75	0.96	0.98
parent-state	0.35	0.75	0.89	0.76

This also settles a question the content study left open, whether a grounded description recovers action as well as a command. Measured from the floor in the lift study, the descriptive ground reaches or exceeds the command in all three of its gating classes (post-write 72 to 66, parent-state 66 to 59, initiative 72 to 71 out of 72; Table 3), and the predicted case where the command does extra work on the smallest model and hardest class did not appear. The one class where the directive edges the descriptive ground is governed, in the separate grounding study above (0.98 vs 0.96). Grounded comprehension alone produces the action.

4.5 Setup versus agency

Recognition without action was measured in a single-turn harness that returns nothing to a tool call, so a model that recognizes a decision has no way to carry it out except by saying so. A minimal tool loop, giving the model a few file and query tools, tests whether the gap is a property of the model or of that setup. Table 6 gives the analysis-mode rate for the glyph condition in raw completion and in the loop. The effect is class-dependent. For classes whose correct action is to act on a known target, reading a state, verifying a result, or consulting a protocol, the loop removes analysis-mode: on parent-state the loop analysis-mode rate is 0, down from 11 in raw completion, and the loop takes the correct action in 13 to 16 of 16 runs; on post-write the rate is 0, down from 14, with correct action in 15 or 16. For classes whose correct action is to assemble a result from several inputs, the gap can survive the loop. The clearest case is formal-step, whose glyph loop rate holds at 7 (and 10 for the imperative). Discovery shows only a small loop rate, 2 for the glyph against a near-zero raw rate, so its evidence is weak, and one class runs the other way: on casg-delegate, whose correct action is to hand off, the loop adds a small analysis-mode rate (4) the raw setup did not; the loop transcripts show the model re-reading files and reaching the step cap without filing the handoff, so this is an execution artifact of the minimal loop rather than analysis-mode. This map rests on one model

and a minimal loop. We offer the split between act-on-a-known-target and assemble-from-inputs decisions as a hypothesis for a fuller test, not an established taxonomy: for the act-on-target classes, recognition without action was largely the single-turn setup denying the model a way to act; for the assemble-from-inputs classes it can reach a minimal agentic setting.

Table 6: Setup versus agency: analysis-mode (Ii) count for the glyph condition, raw completion to tool loop, Qwen3.8-27B, n=16.

Class	raw	loop	reading
parent-state	11	0	removed
post-write	14	0	removed
casg-direct	11	1	removed
formal-step	8	7	persists
discovery	0	2	weak
casg-delegate	0	4	added by loop

One measurement caution governs this comparison. Raw completion credits a stated action, while the loop demands a completed one, so a direct correct-action difference between the two setups mixes analysis-mode with execution failures where a model explores and hits the step cap, a mode consistent with reports that agentic models can overthink rather than act [Cuadron et al., 2025]. The clean cross-setup signal is therefore the analysis-mode rate in the loop, not a raw-minus-loop difference. One blindness limit is specific to this study: a loop transcript and a single-turn reply differ in structure, so a rater can tell the two setups apart. Blindness held to the condition and the predicted direction, and the correct-action bar was identical across setups. This study used one model and a minimal loop, not a full agent harness, and its persistence results are claims about a minimal loop.

4.6 Mechanism typing across the screened set

The controls above run on a few classes. A wider assay types each class by what its suppression or lift depends on. Two damaged versions of a glyph isolate the ingredient: a strongly scrambled glyph keeps the shape and destroys the propositional content, and an off-target glyph keeps coherence and changes the class. A scrambled version that still reproduces the effect means content is not load-bearing; an off-target version that still reproduces it means recognition of the specific class is not load-bearing. Table 7 gives the per-cell counts for the classes that admit clean controls. Comprehension of the content is the load-bearing ingredient in four classes, weakly indicated in a fifth, and one class is mixed across models.

The decision rule reads only cells whose baseline leaves room, and it calibrates on the below-ceiling model with the largest baseline-to-glyph separation, the cell where the effect is cleanest. On that model, a scrambled glyph that reproduces the baseline-to-glyph effect means content is not load-bearing; an off-target glyph that reproduces it means recognition of the specific class is not load-bearing; if both reproduce it, the effect rests on the mere presence of a prefix. By this rule, four classes are comprehension: post-write (all three models, glyph lifts to 8, 20, 22 while scrambled and off-target stay at 0), governed (scrambled 0 against a glyph of 8, 16, 15), structural (Mistral scrambled 6 against glyph 12, baseline 5), and parent-state (phi-4, the largest-separation cell, lifts to a glyph of 20 from baseline 8 while scrambled 9 and off-target 8 stay near baseline; Mistral agrees, scrambled and off-target both 0 against glyph 7). Formal-step points the same way but weakly: only Qwen3.8 leaves room, where a scrambled glyph of 8 stands against a suppressed glyph of 5, a single n=8 cell that does not clear the noise band (Fisher $p = 0.20$), so we read it as provisional. Casg-direct is mixed across the small models: on Mistral, scrambled 1 and off-target 2 reproduce the glyph’s drop to 0, which reads as mere prefix presence, while on phi-4 a scrambled glyph of 6 does not reproduce the glyph’s drop to 2,

Table 7: Mechanism typing: strict-consensus C by class, model, and condition (baseline, glyph, scrambled glyph, off-target glyph), under the strong scramble. Cells are counts out of the per-class pooled total, at 8 runs per model-scenario cell (matching the per-cell n in Table 1): 8 for casg-direct and formal-step (one scenario), 16 for governed and structural (two), and 24 for parent-state and post-write (three). A type is read only on cells where the baseline leaves room; ceiling cells are uninformative.

Class	Model	base	glyph	scram	off
casg-direct	Mistral	8	0	1	2
casg-direct	phi-4	8	2	6	0
casg-direct	Qwen3.8	7	8	8	5
formal-step	Mistral	8	8	8	8
formal-step	phi-4	8	8	8	8
formal-step	Qwen3.8	8	5	8	6
governed	Mistral	1	8	0	1
governed	phi-4	0	16	0	0
governed	Qwen3.8	5	15	0	7
parent-state	Mistral	0	7	0	0
parent-state	phi-4	8	20	9	8
parent-state	Qwen3.8	15	20	15	8
post-write	Mistral	0	8	0	0
post-write	phi-4	0	20	0	0
post-write	Qwen3.8	0	22	0	0
structural	Mistral	5	12	6	5
structural	phi-4	16	16	16	16
structural	Qwen3.8	16	16	14	16

which reads as content-dependent. Neither small-model cell is individually significant ($p = 1.00$ and $p = 0.13$), and the length control (Section 4.3) independently shows a strong generic-length component on this class, so we report casg-direct as mixed with its cleanest cell pointing to mere prefix presence. A type is therefore a joint property of the class and the scramble strength, not of the class alone: under a weaker, keyword-leaking scramble parent-state read as recognition-driven, and the strong scramble reclassifies it to comprehension.

This distribution carries a scope caution it must not outrun. The classes here are the screened set, the taboos we have so far distilled into glyphs and passed through the quality battery. That set is small, and it is drawn from a larger backlog of candidate taboos that was itself assembled in one early period and has not grown since, so it under-samples the decisions we could identify today. The typing is an indication of how the effect distributes, to be tested against a fuller set as more of the backlog is distilled and screened, not a census of a corpus.

4.7 An observer-position constraint

A separate published study bounds what the effect buys. A corpus-loaded assessor and a corpus-naive one reviewed the same agent traces; the naive observer flagged seven items as significant, and the loaded assessor had suppressed all seven, three absorbed and four never seen [Neilson, 2026c]. The sharpest item is an agent that produced correctly structured elimination reasoning while citing evidence that did not support its conclusion: formal compliance without substantive comprehension. This rests on seven items from a single paired comparison, so it is one documented case, not a measured rate. Read that way, it suggests where the present claim should not be pushed: the effect this paper describes is compliance at the level of navigating a decision correctly, which is not the same as performing the reasoning the decision encodes.

4.8 Form can matter, and it is model-dependent

One published result is worth stating plainly, since it can read as a counterexample. On a different task, an investigation with an ordering requirement, the form of the guidance did matter, and how much depended on the model: a prescriptive duty cleared the ordering point in 23 of 24 runs on Qwen2.5-32B where a content-matched descriptive corpus cleared it in 4 of 24, while on Qwen3.8-27B the two forms were indistinguishable and both beat a plain brief, and on Mistral-7B neither beat the brief [Neilson, 2026d]. This is a different contrast, a prescriptive against a descriptive delivery on an ordering task, from the descriptive-against-imperative content match in Section 4.2, and it is reported here alongside rather than as a correction: the form of guidance can matter for some models on some tasks, and the content result says the descriptive glyph form specifically carries no advantage on the decision-execution task. Both results in fact point the same way on mood: the directive delivery does at least as well as the descriptive one, and they differ only in whether that gap is large enough to matter, which depends on the model and the task.

5 Discussion

5.1 What survives

The consolidated picture is that a grounded description of a decision moves a small model to act correctly, through comprehension of the content. The descriptive form is not the lever: a content-matched imperative suppresses and lifts as much (Section 4.2), a length-matched neutral prefix reproduces the terse-class suppression (Section 4.3), and grounding, not the descriptive mood or the named target, is what recovers action (Section 4.4). The effect is class-dependent, and the dependence has a shape: for decisions that act on a known target, recognition-without-action is largely a single-turn artifact that a tool loop removes (Section 4.5), and across the screened set comprehension is the load-bearing ingredient in most classes (Section 4.6). The contribution is this map, not a claim that one form is special. The glyph earns its place as a controlled constant that made the map measurable, and the next stage of the program is a fuller screened set and a form better matched to steering.

The program’s theoretical grounding is the recognition-primed decision account of Klein [Klein, 1998] and the tacit-knowledge account of Polanyi [Polanyi, 1966], in which an agent that recognizes a decision class from prior exposure navigates it without explicit rules. The present results refine that expectation rather than confirm it: in a single-turn setup recognition alone does not produce action, and what converts recognition into execution is grounding the decision in the domain or giving the model a way to act.

5.2 Positioning in the machine-learning literature

The effect sits inside in-context learning, where a prompt conditions behavior at inference time without a weight update [Dong et al., 2023]. It has a training-layer neighbor in Constitutional AI, where a written set of principles shapes behavior through training rather than through a prompt [Bai et al., 2022]; the present result is the inference-time counterpart, a description in context rather than a constitution in the weights. The pragmatic rival to comprehension is implicature: a description may change behavior because a cooperative reader infers an instruction from it, in the sense of Grice [Grice, 1975], and Geng and colleagues [Geng et al., 2026] show social framing shifts instruction-following with content fixed. The present studies do not resolve implicature against comprehension, and Section 5.3 keeps it open. Two cautions from the literature bound the reading of our own measure. A model’s stated reasoning can be unfaithful to the computation behind it [Turpin et al., 2023], which is why the setup study reads a completed action rather than a stated one; and evaluation results can be inflated by

training-data contamination [Golchin and Surdeanu, 2023], which is why the treatment subjects are open-weight models on a local rig rather than a hosted model whose training data is unknown. Finally, the content-over-form result has an adversarial face: if the content an agent reads drives its action, then content an agent merely ingests can carry an injected instruction, the indirect prompt injection surface described by Greshake and colleagues [Greshake et al., 2023]. The program treats that as a separate line of work.

5.3 Limitations

1. **The screened set is small and nested.** The mechanism typing runs on the handful of decision classes distilled into glyphs and passed through the quality battery. That set is a small, processed subset of a larger backlog of candidate taboos, and the backlog was assembled in one early period and has not grown since, so it under-samples the decisions we could now identify. The typing is an indication, not a census. The screening is applied by the author, and which candidate taboos failed it is not enumerated here, so the distribution is conditional on the classes that passed.
2. **Small samples.** Cells are 8 or 16 runs, with a 95% interval of about ± 0.2 and ± 0.12 respectively. We read direction and gaps that clear the band, not single integers.
3. **The domain-free glyph and imperative differ in length.** The descriptive glyph is two to three times longer than the matched imperative (about 940 to 1080 words against 355 to 433). A long prefix depresses small-model action on a terse scenario (Section 4.3), so the imperative’s edge over the glyph in the domain-free contrast cannot be cleanly credited to form; a length-matched imperative would settle it.
4. **A local model shelf.** The subjects are four open-weight models. The findings speak to small language models on this rig, and a different shelf could behave differently.
5. **Single-turn except one study.** All strands but the setup study run single-turn raw completion, and the setup study uses a minimal tool loop, not a full agent harness. The persistence of analysis-mode is a claim about a minimal loop.
6. **Comprehension versus implicature is open.** The studies show content, not form, drives the effect, and that grounding recovers action, but they do not separate a reader comprehending a decision from a reader inferring a command from a description.
7. **Layer invariance is untested.** Whether the effect holds when the description is delivered other than in context, through training or a system prompt or retrieval, is scoped in the program but not yet measured.
8. **The human validity anchor repaired a scoring bug.** The primary raters are blind Claude subagents. To check their validity where it matters most, an author blind-scored a stratified subset of 39 casg-direct responses, weighted to the off-task codes that the length result turns on. The anchor confirmed the off-task coding that carries the length result. It also concentrated the human-versus-machine disagreements at one boundary: whether a reply that asserts or recommends the correct action, without performing it, counts as correct. The first-round rubric under-specified that boundary, and the length-control study had been scored by a single rater, so it credited some asserted or malformed actions as correct. We tightened the rubric and re-scored the four length-control classes blind with two raters and strict consensus, the method the other studies already used, with inter-rater agreement from 0.94 to 1.00. The correction lowered the absolute correct-action rates across all four classes, most on casg-direct and parent-state, where a reply can assert completion, and least on formal-step and conditional-gate, which have a checkable

output. It did not change the load-bearing direction: on the terse class the neutral prefix still suppresses action about as hard as the glyph. One small-cell ordering did move, casg-direct’s neutral prefix against its imperative, and the argument does not rest on it. The off-task mechanism that carries the length result is unchanged (Table 4). The anchor covers one class and one human scorer, and broad validity against a second model family remains future work.

6 Conclusion

A description of a recurring decision, grounded in the concrete domain, moves a small open-weight model to take the correct action, and it does so through comprehension of the content rather than through the descriptive form. A content-matched imperative does as much, a length-matched neutral prefix accounts for the terse-class suppression, and a grounded description that withholds the target still recovers most of the action. Part of what looked like recognition-without-action is a single-turn artifact that a tool loop removes for act-on-target decisions and can leave for assemble-from-inputs decisions, on the one model and minimal loop tested. Across the small screened set of decision classes, comprehension is the load-bearing ingredient in most of them. We report the finding at the scope of the tested models, scenarios, and screened classes, and invite replication on the one local rig the studies run on.

6.1 Data Availability

The study materials, scoring rubrics, per-run scores, and run records for the new studies reported here are available at <https://github.com/sophdn/corpos-lab> under the `studies` directory: `descriptive-material-vs-control`, `neutral-prefix-control`, `grounded-non-prescriptive-aid`, `setup-completion-vs-agentic-loop`, and `alphabet-wide-mechanism-and-grounding-assay`. The published strands cite their own records.

References

- Bai, Y., Kadavath, S., Kundu, S., Askill, A., Kernion, J., Jones, A., Chen, A., Goldie, A., Mirhoseini, A., McKinnon, C., Chen, C., Olsson, C., Olah, C., Hernandez, D., Drain, D., Ganguli, D., Li, D., Tran-Johnson, E., Perez, E., Kerr, J., Mueller, J., Ladish, J., Landau, J., Ndousse, K., Lukosuite, K., Lovitt, L., Sellitto, M., Elhage, N., Schiefer, N., Mercado, N., DasSarma, N., Lasenby, R., Larson, R., Ringer, S., Johnston, S., Kravec, S., El Showk, S., Fort, S., Lanham, T., Telleen-Lawton, T., Conerly, T., Henighan, T., Hume, T., Bowman, S. R., Hatfield-Dodds, Z., Mann, B., Amodei, D., Joseph, N., McCandlish, S., Brown, T., and Kaplan, J. (2022). Constitutional AI: Harmlessness from AI Feedback. *arXiv preprint arXiv:2212.08073*.
- Cuadron, A., Li, D., Ma, W., Wang, X., Wang, Y., Zhuang, S., Liu, S., Schroeder, L. G., Xia, T., Mao, H., Thumiger, N., Desai, A., Stoica, I., Klimovic, A., Neubig, G., and Gonzalez, J. E. (2025). The Danger of Overthinking: Examining the Reasoning-Action Dilemma in Agentic Tasks. *arXiv preprint arXiv:2502.08235*.
- Dong, Q., Li, L., Dai, D., Zheng, C., Ma, J., Li, R., Xia, H., Xu, J., Wu, Z., Liu, T., Chang, B., Sun, X., Li, L., and Sui, Z. (2023). A Survey on In-context Learning. *arXiv preprint arXiv:2301.00234*.
- Geng, Y., Abend, O., Hovy, E., and Frermann, L. (2026). Measuring Pragmatic Influence in Large Language Model Instructions. *arXiv preprint arXiv:2602.21223*.

- Gloaguen, T., Mündler, N., Müller, M., Raychev, V., and Vechev, M. (2026). Evaluating AGENTS.md: Are Repository-Level Context Files Helpful for Coding Agents? *arXiv preprint arXiv:2602.11988*.
- Golchin, S. and Surdeanu, M. (2023). Time Travel in LLMs: Tracing Data Contamination in Large Language Models. *arXiv preprint arXiv:2308.08493*.
- Greshake, K., Abdelnabi, S., Mishra, S., Endres, C., Holz, T., and Fritz, M. (2023). Not what you’ve signed up for: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection. *arXiv preprint arXiv:2302.12173*.
- Grice, H. P. (1975). Logic and Conversation. In *Syntax and Semantics, Volume 3: Speech Acts*, pages 41–58. Academic Press.
- Klein, G. A. (1998). *Sources of Power: How People Make Decisions*. MIT Press.
- Liu, H., Liu, R., Xu, H., Han, J., Fang, S., Yan, S., Deng, H., Wang, G., and Zou, N. (2026). Your LLM, Your Style: Behavioral Mode Axes for LLM Behavioral Control. *arXiv preprint arXiv:2608.10703*.
- McMillan, D. (2026). Instruction Adherence in Coding Agent Configuration Files: A Factorial Study of Four File-Structure Variables. *arXiv preprint arXiv:2605.10039*.
- Neilson, S. D. (2026a). Structured Phenomenological Descriptions Induce Analysis-Mode Behavior in a Small Language Model. *Zenodo*. doi:10.5281/zenodo.22542746.
- Neilson, S. D. (2026b). Content Over Format: What Drives Analysis-Mode in Small Language Models, and How Grounding Recovers Execution. *Zenodo*. doi:10.5281/zenodo.22761018.
- Neilson, S. D. (2026c). Canon Suppression in Corpus-Loaded Assessment: A Two-Assessor Study. *Zenodo*. doi:10.5281/zenodo.22556875.
- Neilson, S. D. (2026d). Duty or Corpus: Whether the Form of Guidance Matters Is Model-Dependent: A Reproducible Single-Turn Behavioral-Equivalence Assay Across Three Models. *Zenodo*. doi:10.5281/zenodo.22716214.
- Pecher, B., Spiegel, M., Belanec, R., and Cegin, J. (2026). Revisiting Prompt Sensitivity in Large Language Models for Text Classification: The Role of Prompt Underspecification. *arXiv preprint arXiv:2602.04297*.
- Polanyi, M. (1966). *The Tacit Dimension*. Doubleday.
- Turpin, M., Michael, J., Perez, E., and Bowman, S. R. (2023). Language Models Don’t Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting. *arXiv preprint arXiv:2305.04388*.
- Wang, Y., Lester, J., and Srivastava, S. (2026). The Story Shapes the Agent: Narrative Priors in LLM Behavior. *arXiv preprint arXiv:2607.18566*.

A Materials

The scenario, glyph, imperative, ground, and neutral prefix for each decision class are reproduced in the repository under the study directories named in Data Availability, with content-parity audits between each aid and its matched control. Representative prompts are reproduced verbatim in the published strands they belong to.